

Name: Solar Panel Team**Date:** 10/10/2025**Class:** Design for Engineers e4usa

MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

This device should be able to operate and swipe over a flat surface.

The design requirement of operating over a solar panel was a key part of Element A, as our previous design attempts lacked this concept. Most designs found improvements on the material side and the inside of the panel rather than the outside. This research provides an opportunity for our design requirement to contribute to a potential design, as previous designs lacked this concept.

The design requirement was approved by the stakeholder. The stakeholder explained that a design solution must reach all areas of the solar panel to be effective.

Design Requirement #2 (HIGH Priority)

The design should be stable and hold itself securely to the panel.

This design ties to Element A because it will allow the solar panels to clean themselves, allowing for higher efficiency. Unlike previous designs, like in Element B, this innovative design will go alongside innovative technology to help increase efficiency to a level where civilians are enticed to buy them.

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

The design requirement was approved by the stakeholder. The stakeholder highlighted the importance of stability, especially on homeowners' roofs or property.

Design Requirement #3 (HIGH Priority)

The design should be able to withstand rough environmental conditions (blizzards, sandstorms, dirt, etc) that can cause weather to affect a solar panel.

This applies directly to prior research from Element A because severe weather has been shown to significantly deteriorate the performance of windshield wipers in cars. So it is important that we create a windshield wiper that does not degrade as much so that it can be used for a long time and does not need to be replaced by homeowners super often.

The design requirement was approved by the stakeholder. The stakeholder mentioned that the more weather-resistant the design is, the more applicable it is as an innovation.

Design Requirement #4 (HIGH Priority)

This device should be able to excrete water and cleaning solutions.

This design requirement applies to element B, as our stakeholder brought up the importance of having a cleaning solution involved in our focus to create a solution that meets our design needs. In other words, element B's recommendations allow us to facilitate our project goals through the design requirement of providing necessary cleaning solutions.

The design requirement was approved by the stakeholder. The stakeholder mentioned that perfecting our cleaning solution was important if we were going to create a design



Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

around it.

Design Requirement #5 (HIGH Priority)

This device should be able to operate (come into contact with the panel) without damaging the panel itself.

In our discussions with our stakeholders in Element B, a common theme of operating smoothly came up in all designs. This is critical in providing a uniform solution in our design that can meet the needs of the stakeholders in element B.

The design requirement was approved by the stakeholder. Our stakeholder mentioned that having our device last longer is key to success.

Design Requirement #6 (HIGH Priority)

The design should improve the solar panel efficiency.

This applies directly to Element A because we found that solar panel efficiency is increasing, but it has been a very slow increase every year. As a result, the product will help improve the effectiveness of solar panels by cleaning them more often and outputting more solar energy.

The design requirement was approved by the stakeholder. Our stakeholder recognizes that solar panel efficiency is key to improving and innovating on previous designs.



Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

Design Requirement #7 (LOW Priority)

The design should be creative and aesthetic.

This applies directly to prior Element A design research because previous economic studies have found that people tend to buy things that look cool since they catch their attention. Additionally, homeowners might not want an unpleasant-looking solar panel on their house.

The design requirement was approved by the stakeholder. Our stakeholder added that solar panels that look good can add to the property value of a house, which is an additional benefit to our design.